

Dylan T. Jacobs

Curriculum Vitae

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Swarthmore College
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EDUCATION

Swarthmore College

B.S. General Engineering (electrical focus), B.A Applied Mathematics

GPA: 3.98/4.00

Relevant Coursework: Electronic Circuit Applications, Digital Signal Processing, Electrodynamics, Computer Vision, Data Structures & Algorithms, Partial Differential Equations, Numerical PDEs, Statistical Signal Processing

Swarthmore, PA

2023–2027 (expected)

AREAS OF INTEREST

- Physics-informed neural networks (PINNs)
- Biomimetic and soft robots
- Control theory and reachability for autonomous systems
- Hyperbolic conservation laws and transport phenomena
- Kinetic modeling and Eulerian-Lagrangian methods

PUBLICATIONS

Pre-prints

[1] J. Nakao, **D.T. Jacobs**, and W. Taitano, "An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system", *Submitted* (2026).

INTERNSHIP AND RESEARCH EXPERIENCE

US Naval Research Laboratory

Power Electronics Division

Washington, DC

05/2026–08/2026

Mentors: Benjamin Greenberg and Boris Feigelson

Project 1: Developing inductively coupled plasma (ICP) reactor to synthesize nanoparticles and conducting ICP experiments.

- Minimized reflected power in ICP by tuning the RF matching network using Smith chart.
- Mitigated RF electromagnetic interference by designing Faraday cage.

Project 2: Scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDXS) of atomic-layer-deposition coated nanomaterials.

- Designed and ran atomic-layer-deposition (ALD) experiments to determine how dose parameters affect ALD coatings.

- Performed SEM and EDXS to characterize ALD-coated compacts and inform experimental design.

US Naval Research Laboratory

Power Electronics Division

Washington, DC

05/2025–08/2025

Mentor: Benjamin Greenberg

Project 1: Designing automated liquid-nitrogen (LN₂) delivery system to cool laser flash analysis (LFA) instrument.

- Designed and built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring to automate LFA experiments, including a GUI and data-logging interface to the lab computer.

Project 2: Characterizing thermoelectric nanomaterials using laser flash analysis (LFA).

- Measured thermal diffusivity, specific heat, and Seebeck coefficients of thermoelectric semiconductors using LFA and Seebeck Analysis.
- Improved the measurement procedure for infrared-transparent and high-resistance samples.

Swarthmore College NaSCaRS Lab

Applied Mathematics Department

Swarthmore, PA

01/2024–present

Mentor: Joseph Nakao

Project 1: Low-rank, structure-preserving solver for 1D2V Vlasov-Fokker-Planck plasma equation in cylindrical coordinates.

- Presented posters at 2025 SIAM NNP conference, 2024 Sigma Xi Swarthmore poster session.
- Giving talk at upcoming SIAM NNP 2026 conference.

Current Project: Eulerian-Lagrangian method with CUR decomposition for solving advection-diffusion equations in C++

Swarthmore College Masroor Lab

Engineering Department

Swarthmore, PA

01/2024–05/2024

Mentor: Emad Masroor

Project: Developing small-scale, oscillatory wind-energy harvesting device.

Oregon Health and Science University

Artificial Intelligence for Medical Systems Lab

Portland, OR

06/2022–08/2022

Mentor: Clara Mosquera-Lopez

Project: Coding Android app in Kotlin to detect heart murmurs from Bluetooth audio data.

SCHOLARSHIPS AND AWARDS

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| • Clearfield ASHE Scholarship, <i>American Society of Highway Engineers</i> | 04/2026 |
| • Delaware Valley Engineers Undergraduate Scholarship, <i>Delaware Valley Engineers Society</i> | 02/2025 |
| • Delaware Valley Engineers Undergraduate Scholarship, <i>Delaware Valley Engineers Society</i> | 02/2025 |
| • Donna Prentice Memorial Scholarship, <i>American Society of Civil Engineers</i> | 02/2024 |
| • National Merit Scholarship | 04/2023 |
| • Global Seal of Bilingualism (Spanish) | 07/2022 |

CONFERENCE TALKS AND POSTERS

Conference Talks

- An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system 10/2026 (upcoming)
SIAM NNP Section Conference, Rutgers University, New Brunswick, New Brunswick, NJ

Posters

- Plasma Synthesis and Scanning Electron Microscopy Characterization of Nanomaterials 09/2026
Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA
- An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system 11/2025
SIAM NNP Section Conference, Penn State University, State College, PA
- Laser Flash Analysis Characterization of Thermoelectric Nanomaterials 09/2025
Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA
- A Robust, Third-Order Accurate Solver for the 1D1V Vlasov-Poisson Equation 09/2024
Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA

Conferences Attended

- SIAM CSE 2026 02/2027 (upcoming)
Pittsburgh, PA
- SIAM NNP Section Conference 2026 10/2026 (upcoming)
Penn State University, State College, PA
- SIAM CSE 2025 03/2025
Fort-Worth, TX
- SIAM NNP Section Conference 2025 11/2025
Penn State University, State College, PA

SELECTED PROJECTS

Convolutional Neural Network for Hand-Drawn Image Classification: [link](#) 05/2026

Designed and trained a CNN to classify 250 object categories, deployed it online for real-time evaluation using ONNX.

Component-Wise Kalman Filter Model for Smartphone Battery Drain: [link](#) 01/2026
COMAP Mathematical Competition in Modeling (MCM)

Developed a battery state-space model with Kalman filtering using a system of ODEs, combined with Coulomb-Counting and battery Equivalent Circuit Model (ECM), to predict smartphone battery time-to-empty given time-varying operating and use conditions.

Analog Circuit and PCB Design 08/2025 — 12/2025

Designed and fabricated PCBs in Fusion360 combining MOSFETs, sensors, actuators, op-amps, comparators, and audio amplifiers across four projects: [analog audio equalizer](#), [temperature-controlled fan](#), [light-optimizing solar panel](#), [regulated DC power supply](#).

AI Python Stock Trading Algorithm: [link](#) 02/2025

Trained neural network on technical indicators to trade stocks using Alpaca API.

Generative Adversarial Network (GAN): [link](#) 02/2023

Image-generating model trained on web-scraped art.

TEACHING ASSISTANT ROLES

Swarthmore College

Teaching Assistant for Electrical Circuit Analysis

Swarthmore, PA

08/2025–12/2025

- Led tutoring, problem-solving, and review sessions for Swarthmore engineering students.
- Helped students develop problem-solving strategies, as well as foundations in mathematics, electrical engineering, and signal processing to solve circuit analysis problems.

Swarthmore College

Mathematics Tutor

Swarthmore, PA

08/2025–present

- Leading one-on-one tutoring sessions for Swarthmore mathematics students in a wide range of courses, from Calculus I, II, III to Differential Equations, to Real Analysis.

PROFESSIONAL SERVICE

Memberships

- National Society of Professional Engineers
Member. 09/2026–present
- Tau Beta Pi Engineering Honor Society
Member of the Swarthmore College Chapter of the Tau Beta Pi Engineering Honor Society. 02/2026–present
- Sigma Xi
Member of the Swarthmore College Chapter of Sigma Xi. 08/2024–present

Extracurriculars

- Swarthmore Jazz Ensemble 08/2026–present
- Swarthmore Men's Varsity Soccer 08/2023–06/2026
- Swarthmore College Computer Society 01/2024–05-2024

SKILLS

Hardware: SPICE, PCB Design, CAD, Fusion360, Soldering, Embedded systems

Software: TensorFlow, PyTorch, OpenCV, Git, $\text{\LaTeX}$

Programming: Python, Arduino, MATLAB, C/C++,

Languages: Spanish (fluent)